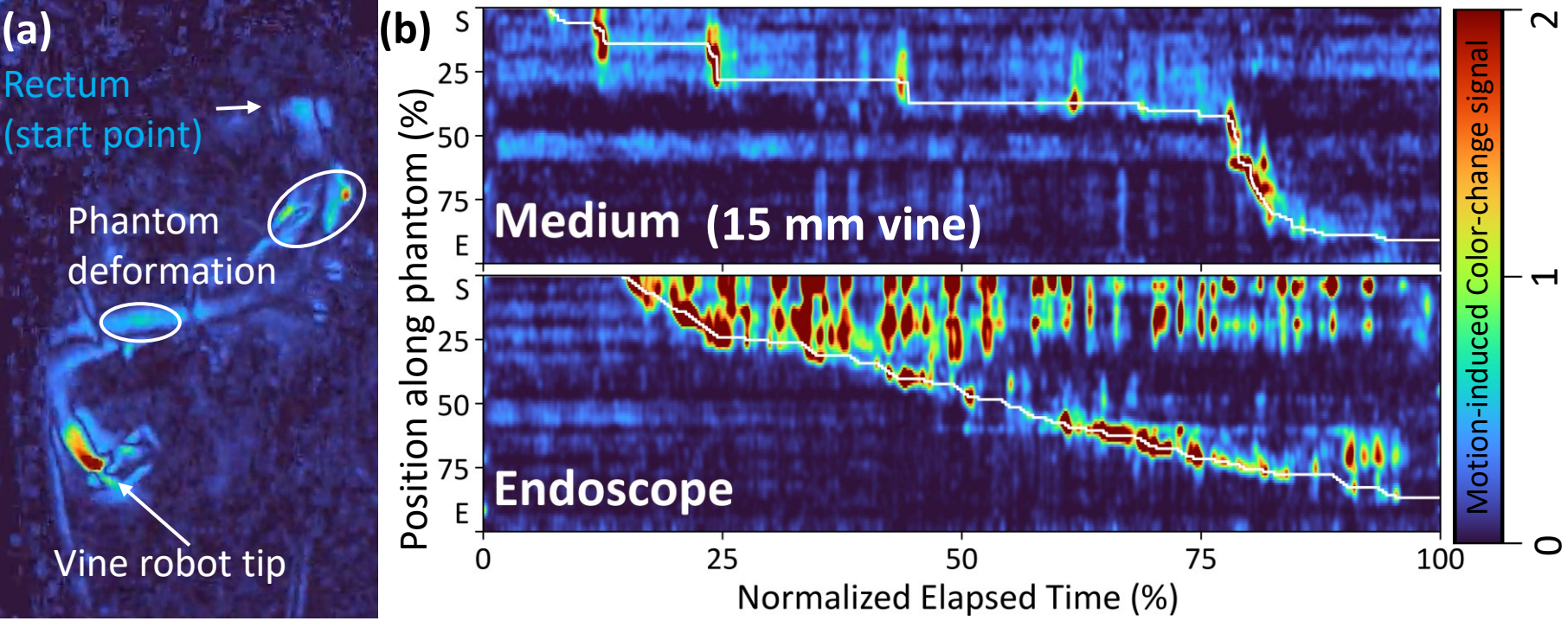

(a)
Rectum
(start point)
Phantom
deformation
Vine robot tip
(b)
Medium (15 mm vine)
Endoscope
Position along phantom (%)
S
25
50
75
E
0
25
50
75
100
Normalized Elapsed Time (%)
Motion-induced Color-change signal
0
1
2